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Github:- https://github.com/amitishusharma2005-prog/CS3081-Kd-command-pons

1. What I learned

A small student model (Pythia-70M) can be trained on the text that a bigger teacher (Phi-3.5) generates, this is a sequence-level distillation.
I learned to split work into chords, to save checkpoints so a crash does not lose work, to keep liveness and readiness probes separate so a pod that is still loading gets no traffic and that a network ACL is stateless.

2. Challenges and how I solved them

- The teacher ran only on the CPU because the GPU failed so labelling 3000 commands took many hours. Ollama hung for about nine hours and the storage had an outage. Workers save a checkpoint every 50 commands and resume; I added retries.
I changed the teacher request options part-way (about the first 1700 replies came before). I wrote this in the report
- My first deployment test failed: 99.8 % load test error, 14.7 %. rolling-updates and 32.8 % recovery failures, and after rollback V2 still answered.
The 1 CPU limit starved the health probes. I made each pod to do one generation at a time, lengthened

probe timeouts and the shutdown path, fixed the rollback check, and added a warm-up before a pod is ready. Rolling update failure went 2.15%, 3.33%, 1.50%, then 0%.
One CI lint job failed because of a long line; I fixed it.

3. What I measured
Cleaning kept 2863 of 3000; I trained on 1500. Validation (100 commands), JSON validity and full match: teacher 99% and 48%, student v1 100% and 53%, student v2 100% and 48%, untrained 0% and 0%.
load test: 0% error, p95 2.45. Rolling update: 0 of 469 failed, rollback shows v1. Recovery 0 of 351 failed. HPA went 1 to 2 and back to 1. Replicas served 56% and 44%. Cold start median 685. Cost: 0.8165 INR per 1000 request.

4. Observation and Improvement
The student is only as good as its labels, because teacher mistakes that took valid are copies. One laptop cannot show fault tolerance across machine. Model loading dominates cold start. I would use a larger student, batching or quantisation, a multi-node cluster, ~~shot~~ short lived credentials with TLS, and a GPU for the teacher.